

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q16-0596

North Charlotte 100 kV
April 16, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0596. The Interconnection Customer proposes a 161.2 MW battery energy storage facility interconnecting at the North Charlotte 100 kV substation in the Illinois area. The request was received on March 11, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 196 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 11 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$3,962,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Ege - Cuba City 100.0 kV
Contingency: Dekalb - Jones Chapel Substation 500.0 kV
Loading: 128.1% (Rating: 170 MVA)
DFAX: 5.4%
Pre-project loading: 123.0%
2. Facility: Ege - Goldn Gate 100.0 kV
Contingency: Dekalb - Jones Chapel 500.0 kV
Loading: 124.5% (Rating: 185 MVA)
DFAX: 5.4%
Pre-project loading: 119.8%
3. Facility: Laureles - Ward Gulf Substation 100.0 kV
Contingency: Dekalb - Jones Chapel 500.0 kV
Loading: 114.6% (Rating: 281 MVA)
DFAX: 5.4%
Pre-project loading: 111.5%
4. Facility: North Charlotte - Navy 1 100.0 kV
Contingency: Magnesium Plant - Navy 1 100.0 kV
Loading: 109.3% (Rating: 136 MVA)
DFAX: 21.4%
Pre-project loading: 83.9%
5. Facility: Laotto - GoldenGate 100.0 kV line
Contingency: Dekalb - Jones Chapel Substation 500.0 kV
Loading: 108.5% (Rating: 226 MVA)
DFAX: 5.4%
Pre-project loading: 104.7%

6. Facility: Laureles - Big Oaks 100.0 kV
Contingency: Dekalb - Jones Chapel 500.0 kV
Loading: 107.4% (Rating: 321 MVA)
DFAX: 5.4%
Pre-project loading: 104.7%
7. Facility: Big Oaks - Cypress Creek 100.0 kV
Contingency: Jones Chapel - Dekalb 500.0 kV Line
Loading: 105.2% (Rating: 349 MVA)
DFAX: 5.4%
Pre-project loading: 102.7%
8. Facility: Cypress Crk - Ligonier Substation 100.0 kV
Contingency: Jones Chapel - Dekalb 500.0 kV
Loading: 104.3% (Rating: 360 MVA)
DFAX: 5.4%
Pre-project loading: 101.9%
9. Facility: Dairymans - Pueblo Xtra 230.0 kV Circuit 2
Contingency: Pueblo Xtra - Dairymans 230.0 kV
Loading: 104.1% (Rating: 846 MVA)
DFAX: 9.2%
Pre-project loading: 102.3%
10. Facility: Baraga Vlg 500-230 kV Transformer
Contingency: Wolfe - Barrett 500.0 kV
Loading: 103.8% (Rating: 1,039 MVA)
DFAX: 8.5%
Pre-project loading: 102.5%
11. Facility: Stapley - Rosemore B 100.0 kV L
Contingency: Stapley - Rosemore B 100.0 kV #1
Loading: 101.1% (Rating: 159 MVA)
DFAX: 12.5%
Pre-project loading: 88.5%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Ege - Cuba City 100.0 kV	Rebuild line with higher-capacity conductor	\$6,281,000	3% (\$166,000)
Ege - Golden Gate 100.0 kV	Rebuild line with higher-capacity conductor	\$2,223,000	3% (\$59,000)
Ward Gulf - Laureles Substation 100.0 kV	Rebuild line with higher-capacity conductor	\$4,312,000	3% (\$114,000)
North Charlotte - Navy 1 100.0 kV	Reconductor line	\$841,000	100% (\$841,000)
Laotto - GoldenGate 100.0 kV line	Reconductor line	\$1,542,000	3% (\$41,000)
Big Oaks - Laureles 100.0 kV L	Reconductor line	\$1,468,000	3% (\$39,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Big Oaks - Cypress Crk Substation 100.0 kV	Reconductor line	\$2,262,000	3% (\$60,000)
Ligonier - Cypress Crk 100.0 kV L	Reconductor line	\$3,280,000	3% (\$87,000)
Dairymans - Pueblo Xtra 230.0 kV Circuit 2	Reconductor line	\$24,088,000	4% (\$911,000)
Baraga Vlg 500-230 kV Transformer	Replace transformer with higher-rated unit	\$30,837,000	3% (\$847,000)
Stapley - Rosemore B 100.0 kV L	Reconductor line	\$797,000	100% (\$797,000)
	Total allocated to project		\$3,962,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.